Clinical Intelligence & Data Strategy

# COVID-19 vs.
# Respiratory Pathologies

Simulating Real-World BI Challenges with Messy Data

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# Project Objective & Methodology

### Data Cleaning

Cleaning messy raw data from GitHub to simulate real-world BI challenges and system errors.

### Clinical Insights

Finding  insights for hospital management by comparing pathologies side-by-side.

### Modern Tools

**Microsoft Power BI** and **Power Query** for end-to-end transformation.

# Critical Data Quality Challenges

**Broken Timelines:** Missing records for female patients in temperature and oxygen charts.

**PCR Trap:** Incomplete PCR results required filtering by clinical 'Findings' instead.

**Impossible Metrics:** Outliers (Temp > 40°C) were converted to nulls to protect averages.

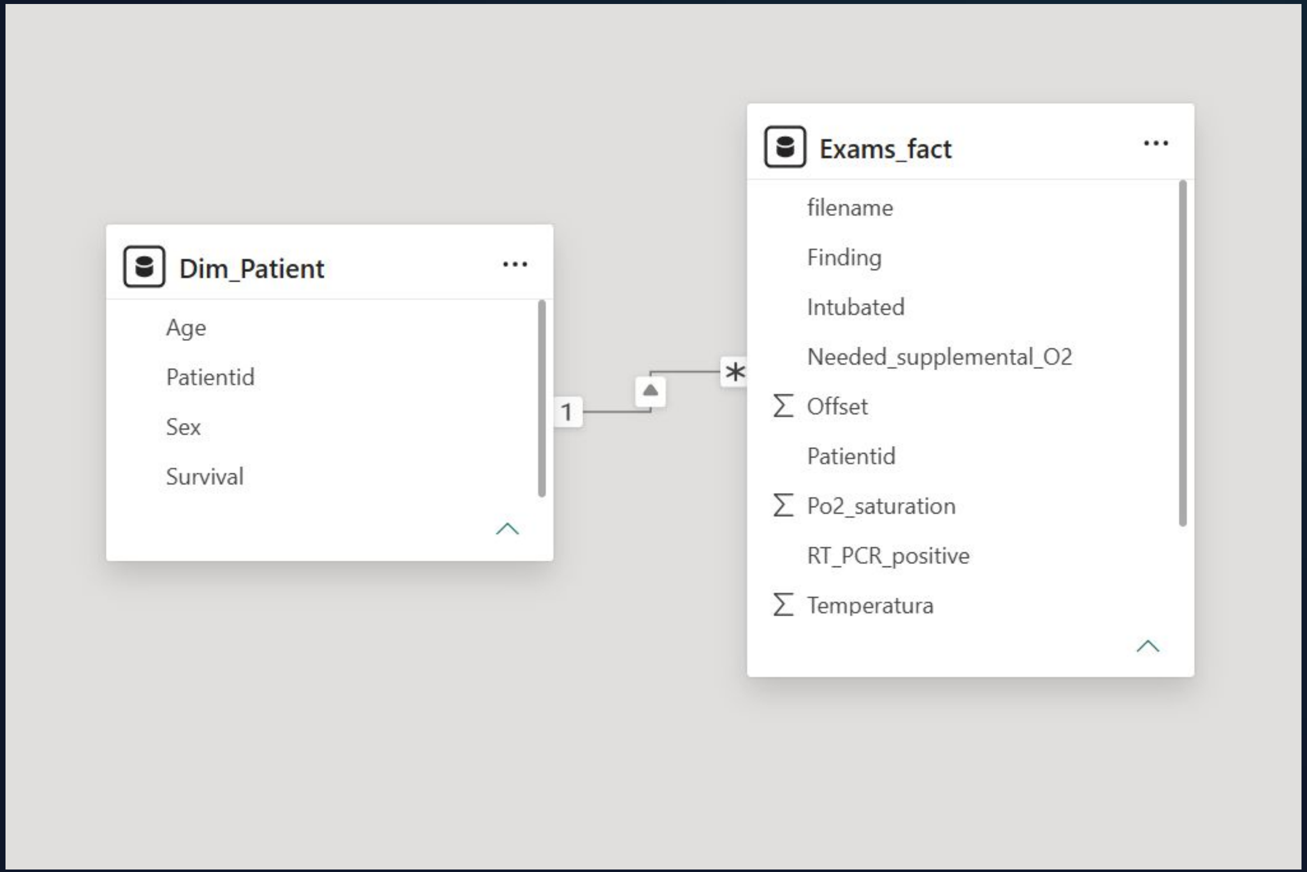
| | ABC 123 Column1 | ABC 123 Column2 | ABC 123 Column3 | ABC 123 Column4 | ABC 123 Column5 | ABC 123 Column6 |
|---|---|---|---|---|---|---|
| 1 | patientid | offset | sex | age | finding | RT_PCR_positive |
| 2 | 2 | 0 | M | 65 | Pneumonia/Viral/COVID-19 | Y |
| 3 | 2 | 3 | M | 65 | Pneumonia/Viral/COVID-19 | Y |
| 4 | 2 | 5 | M | 65 | Pneumonia/Viral/COVID-19 | Y |
| 5 | 2 | 6 | M | 65 | Pneumonia/Viral/COVID-19 | Y |
| 6 | 4 | 0 | F | 52 | Pneumonia/Viral/COVID-19 | Y |
| 7 | 4 | 5 | F | 52 | Pneumonia/Viral/COVID-19 | Y |
| 8 | 5 | null | null | null | Pneumonia | null |
| 9 | 6 | 0 | null | null | Pneumonia/Viral/COVID-19 | Y |
| 10 | 6 | 4 | null | null | Pneumonia/Viral/COVID-19 | Y |
| 11 | 3 | 4 | M | 74 | Pneumonia/Viral/SARS | null |
| 12 | 3 | 9 | M | 74 | Pneumonia/Viral/SARS | null |
| 13 | 3 | 10 | M | 74 | Pneumonia/Viral/SARS | null |
| 14 | 7 | 7 | F | 29 | Pneumonia/Viral/SARS | null |
| 15 | 7 | 12 | F | 29 | Pneumonia/Viral/SARS | null |
| 16 | 8 | 9 | F | 42 | Pneumonia/Viral/SARS | null |
| 17 | 9 | 5 | F | 46 | Pneumonia/Viral/SARS | null |
| 18 | 9 | 17 | F | 46 | Pneumonia/Viral/SARS | null |
| 19 | 10 | 19 | F | 73 | Pneumonia/Viral/SARS | null |
| 20 | 10 | 27 | F | 73 | Pneumonia/Viral/SARS | null |
| 21 | 10 | 35 | F | 73 | Pneumonia/Viral/SARS | null |

# Strategic Data Modeling

## The Star Schema Advantage

To ensure efficiency, we transformed the flat spreadsheet into a relational model:

- **Patient Dim:** Unique demographics (Sex, Age, ID).
- **Exams Fact:** Multi-event clinical data (X-rays, Vital signs).
- **Outcome:** Faster queries and distinct patient filtering.



# Transformation Logic - Step 1

## ID & Timeline Fixes

Cleared text noise from IDs (e.g., "311.a" to 311). Handled duplicate IDs by linking them to unique **Offset** events to track disease progression.

## Data interval Limits

Implemented strict filters: Maximum temperatures at 40°C and O2 saturation at 100%. Anything above or under zero became **null**.

# Standardizing Clinical Status

**Demographics:** Mapped "M/F" to "Male/Female" and unknowns to "Not Informed".

**Critical Status:** Mapped "Y/N" for survival, intubation, and ICU status to clean "Yes/No" labels.

**Noise Removal:** Technical blood metrics (WBC, Neutrophils) were removed to prioritize readable hospital management KPIs.

# Visual Integrity & Symmetry

**1:1**
AXIS SYMMETRY

**457**
UNIQUE PATIENTS

Locked axes and synchronized timelines ensure that managers can compare pathologies **visually** without scaling errors.

# Patient Demographic Breakdown


Clinical Significance: Men show higher hospitalization and complication rates globally.

# The COVID-19 "Rollercoaster"


## Temperature Patterns

**Days 1-5 (Viral):** High initial fever.

**Day 6 (False Recovery):** Temperature drops to 36/37° C.

**Day 10+ (Inflammation):** Sudden spike to 39°C.

This identifies the critical phase where aggressive intervention is needed.

# Oxygen Drops & ICU Peaks

COVID-19 patients experience a drastic crash in oxygen levels between **Days 12-15**.

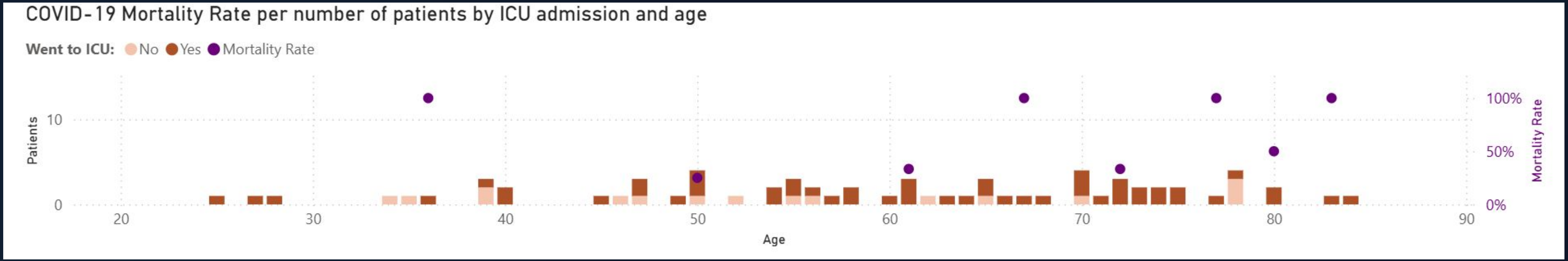
- **Late Arrival:** Intubations peak at Day 0 (54 cases).
- **Reason:** Isolation at home causes patients to arrive with destroyed lungs.
- **Survival:** Ventilators were critical for saving 71% of ICU patients.

# Age & Biological Fatal Limits

## Mortality Rate

Analysis of over 61-year-olds showed **every single patient** who died was already in the ICU.

**Conclusion:** On this dataset the high mortality in the elderly is mainly for biological reasons and not due the lack of ICU.



# Any Questions?

Strategic Insights for Hospital Management through Data Quality.

$$\text{Survival Rate} = \frac{\text{Survived}}{\text{Total Patients}} \times 100$$

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**Source: GitHub COVID Dataset**

https://raw.githubusercontent.com/ieee8023/covid-chestxray-dataset/master/metadata.csv)